fire and buried their dead. They did not practice agriculture and animal domestication.

Cro-Magnon was one of the most talked forms of modern human found from the rocks of Cro-Magnon, France and is considered as the ancestor of modern Europeans. They were not only adapted to various environmental conditions, but were also known for their cave paintings, figures on floors and walls.

Homo sapiens or modern human arose in Africa some 25,000 years ago and moved to other continents and developed into distinct races. They had a brain capacity of 1300 – 1600 cc. They started cultivating crops and domesticating animals.

6.9 Isolating Mechanism

Isolation is the separation of the members of a single population into sub populations so that genetic integrity of the subpopulation can be maintained. Closely related species living in the same area do not breed together; they are prevented by isolating barriers. An isolating barrier is any evolved character of the two species that stops them from interbreeding. Several kinds of isolating barriers are distinguished. The most important distinction is Prezygotic and post zygotic isolation. Prezygotic mechanisms include those which prevent two species from coming into contact. This includes ecological, seasonal, ethological and morphological. Post zygotic mechanisms are those which act after fertilization that include hybrid sterility, hybrid inviability and hybrid breakdown.

Pre-zygotic Mechanism

i. **Ecological isolation or habitat isolation** – the members of the same population may be separated from one another by a differences in their habitat. For example *Rana areolata* occupies burrows dug by mammals and tortoises during the day and breeds in grassy shallow ponds whereas *Rana grylio* breeds in deep waters. Due to the difference in their habitat the two species are able to maintain their respective species identities.

ii. **Seasonal isolation** – In this type of isolation, difference in the breeding seasons prevents interbreeding. E.g. Toad, *Bufo americanus* breeds much early in the spring; whereas *Bufo fowleri* breeds very late in the season. They are able to maintain their species identity because of the differences in the breeding seasons.

iii. **Sexual or ethological isolation/ Behavioural isolation** – Prevents mating due to the difference in their sexual behavior. The species are not separated from one another either in time or in space. The mating calls of two closely related species of frogs, *Hyla versicolor* (grey tree frog) and *Hyla femoralis* (pine wood tree frog) are different which prevents interbreeding.

iv. **Morphological isolation or mechanical isolation** – This type of isolation is due to the differences in their external genitalia that is seen in two different species. The size difference between the toad species *Bufo quercicus* and *Bufo valliceps*, prevents their interbreeding.

v. **Physiological isolation** –Though mating may occur, the gametes are prevented from fertilization due to mechanical or physiological factors. E.g. The sperms of *Drosophila virilis* survive only for about a day when introduced into the sperm receptacle of *Drosophila americana* while the sperms of *Drosophila americana* live for a longer time.

vi. **Cytological isolation** – Fertilization does not take place due to the differences in the chromosome numbers between the two species, the bull frog *Rana catesbiana* and gopher frog *Rana areolata*.

Post-zygotic Mechanism

i. **Hybrid inviability** – In this type, the sperm enters the egg, fertilization occurs and the embryo develops into the adult but it dies before reaching maturity. In certain fishes,

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